

is normalisation. When the 'gausssize' filter option is set at the lower end of 3, it behaves like a typical compressor. At the other end of 300, it becomes a traditional normaliser.

```
ffmpeg -i train-trip.mp3 \
    -filter:a dynaudnorm=gausssize=3
\
    train-trip-normalized.mp3
```

Do not use the filter indiscriminately. *Dynamic audio compression* is also known as *dynamic range compression*. It is the bane of popular music today. It makes the recording very boring. In Carl Orff's composition of *O Fortuna* or Ryuichi Sakamoto's score for the end-credits of the movie *Femme Fatale*, the music starts on a low note, builds slowly in a steady crescendo and abruptly drops off a high cliff. Compressing such an audio will ruin the composer's intent.

Using FFplay and 'dynaudnorm'

Despite having a fibre internet connection, its challenging to play online videos on my PC, let alone stream them to my TV. To address this, I have written a browser script (www.opensourceforu.com/2016/03/the-utility-of-user-scripts-js-and-user-styles-css) that automatically detects online videos and displays their location (URL). Using this script and a download manager, I mass-download videos and see them offline.

Another problem with the videos (recorded mostly by other writers) is that they have very loud intro music and the useful content is recorded with low volume. Rather than fiddle with volume control for every video, I use a *Caja Action* that uses the 'dynaudnorm' filter and 'ffplay'. In addition to files, 'ffplay' can play the processed output of filters in real-time. I use the '-f lavfi' option for this.

```
sFile=Self-Editing-Tips-from-an-Editor.
mp4
```

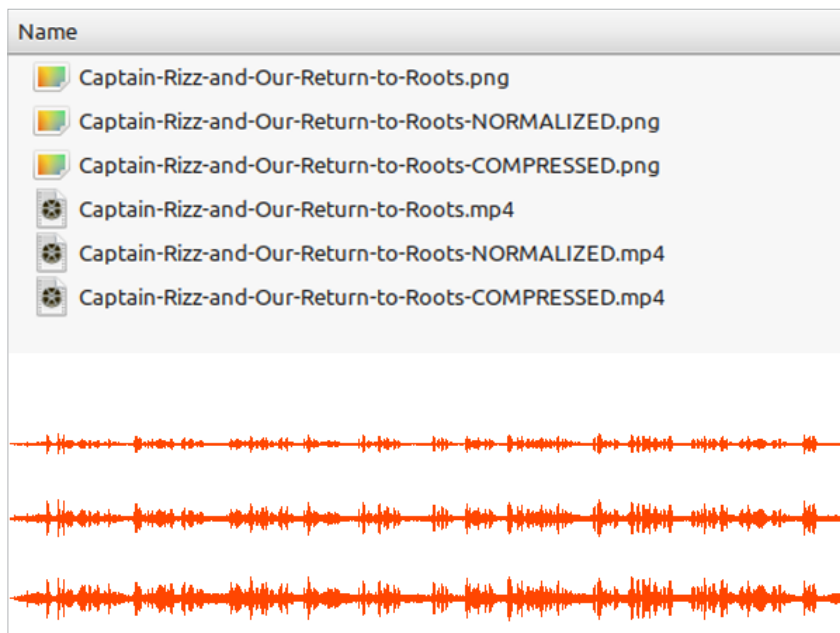


Figure 3: Waveform images of an audio file before and after normalisation and compression

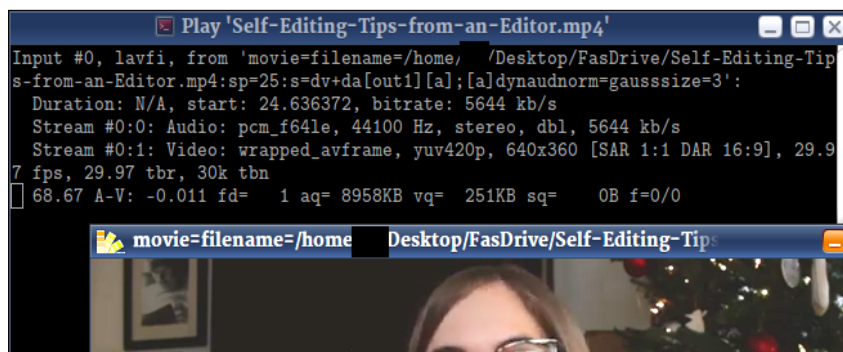


Figure 4: FFplay can play the output of filters in real-time

```
ffplay -hide_banner -autoexit -f lavfi \
    "movie=filename=${sFile}:sp=25:s=dv+da[out1][a];
    [a]dynaudnorm=gausssize=3"
```

The 'movie' filter plays the video. Its 'sp' option skips the first 25 seconds. The 'dynaudnorm' dynamically compresses the remaining audio.

I am listening to other authors talking about their books. It is not

classical music or some symphony orchestra. Compressing such videos is entirely justified.

Some online video creators use a microphone with an active compressor. Their videos have a constant volume. For content creators who mix intro or outro music, I recommend processing their videos with the 'dynaudnorm' filter before uploading the output. END 🐧

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